Gauge Structure, QED, and Gravity from the Spectral Geometry of S^3: A Synthesis of the GeoVac QED/Gauge Arc*

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The Geometric Vacuum (GeoVac) framework labels the hydrogen bound states by the harmonics of the unit three-sphere S^3 via Fock's conformal projection. This synthesis assembles the eight Group 5 papers—gauge structure, QED spectral sums on S^3 and the graph, precision bound-state QED, and a spectral-action gravity sketch—into one narrative around a single organizing distinction: *$U(1)$ and* SU(2) *Wilson structures can be built on the graph, the choice of* SU(2) *being natural because $S^3 =$* SU(2)*, while the couplings are calibration/observation, not derived.* The graph carries a $U(1)$ Wilson/Hodge lattice gauge structure [6], a non-abelian SU(2) Wilson theory because $S^3 =$ SU(2) as a manifold [8], and—on the cross-ℓ Dirac graph—a $U(1)$ theory compatible with compact electrodynamics, with no 3D-versus-4D discrimination [14]; the natural gauge reach is $U(1)\times$SU(2)$\times$ SU(3) (an organizing reading, not a derivation), with SU(3) entering only as a pure-gauge theory whose natural matter coupling is obstructed. Only one of eight selection rules (Gaunt sparsity) is graph-native; the rest are scored in a vertex model with vector-photon labels, whose ground-state rule is a scalar-triangle zero and whose eighth rule rests on an α_z identity that a physical vector photon does not satisfy [10]; the period content of the model's spectral sums enters through the continuum S^3 Dirac sums, which produce the odd zeta values of flat-space QED ($\zeta(3)$, $\zeta(5)$), plus S^3 vertex-parity content (Catalan G, $\beta(4)$) and the depth-2 period $\zeta(5,3)$, through a depth-filtered period tower [7]. The fine-structure constant approximately satisfies (to 8.8×10^{-8} relative; an Observation) the cubic $\alpha^3 - K\alpha + 1 = 0$ with $K = \pi(B + F - \Delta)$ assembled from three spectral invariants of the Hopf bundle, each with an independently established spectral home; this is recorded strictly as an **Observation** (agreement 8.8×10^{-8}, twelve single-mechanism derivations eliminated), never as a derivation [2]. The framework's Bethe-logarithm machinery reproduces the tabulated hydrogen values for $2P$ and $3D$, with the $2S$ value not yet converged; the sub-percent Lamb-shift headline is the textbook leading one-loop assembly with literature logarithms [12]. A spectral-action gravity sketch uses the two-term Einstein–Hilbert action of the round sphere (Chamseddine–Connes; an alternative proof here), together with a set of honest negatives that bound a still-open discrete-substrate program [15]. The discipline throughout matches the rest of the corpus: every load-bearing claim wears its tier, the structural results are foregrounded, and the couplings are positioned as the free, calibration side of the construction.

I. INTRODUCTION: NATURAL STRUCTURE, FREE COUPLINGS

The GeoVac program builds its discrete hydrogen lattice on the labels (n, l, m) of the Laplace–Beltrami operator on the unit three-sphere S^3, on which that operator acts as the diagonal $SO(4)$ Casimir; the operator is related to the Schrödinger equation by Fock's 1935 conformal projection [3]. (The lattice's graph Laplacian is not equivalent to it and does not converge to it, Paper 7.) The Group 5 arc asks what the *same* spectral geometry says about the rest of electromagnetism, the gauge sector, and—most speculatively—gravity. A single distinction organizes the answer, and it is the through-line of this synthesis.

The $U(1)$ and SU(2) *gauge structures are constructions the geometry makes natural, not consequences it forces; the $U(1) \times$ SU(2) $\times$ SU(3) saturation below is an organizing reading, not a derivation.* The three-sphere carries a canonical principal-bundle structure, the Hopf fibration $S^1 \to S^3 \to S^2$, and S^3 *is* the group manifold SU(2). These two facts are not GeoVac inputs; they are geometry. Reading the GeoVac graph through its incidence matrix, one can build an abelian and then a non-abelian Wilson lattice gauge theory on the same nodes and edges (a Wilson theory can be placed on any graph; what the geometry supplies is the natural choice of group), whose labels carry the Coulomb spectrum (§II; the spectrum sits on the labels as the diagonal $SO(4)$ Casimir, not in the edge Laplacian). The natural gauge reach saturates at $U(1) \times$ SU(2) $\times$ SU(3), and where it stops is itself structural: SU(3) admits only a pure-gauge Wilson theory, its natural matter coupling obstructed at the Clebsch–Gordan intertwiner level (a sharpening of the Coulomb/harmonic-oscillator asymmetry of Paper 24 [5]).

The couplings are not. The fine-structure constant, the QED multi-loop constants, and the cutoff function of the gravity sector are all on the other side of the ledger: calibration or observation, in the two-layer language of Paper 34 [11]. The framework's most precise number—the cubic's root matches α to 8.8×10^{-8} (K itself to 4.8×10^{-7})—is held explicitly at **Observation** tier, with no claim of derivation (§IV). This is the same discipline that runs through the quantum-chemistry and quantum-

* Group 5 synthesis in the Geometric Vacuum series

computing arcs: lead with the structural result, name the calibration boundary, and let the tier of each claim be visible in the prose.

Following the framework's positioning convention, the discrete graph topology and the continuous Hopf/spectral-action geometry are treated as *dual descriptions* connected by proven maps; no ontological priority is asserted for either, and the synthesis leads with concrete structural content rather than interpretive claims about the nature of electromagnetism or gravity.

The eight papers form a loose dependency order. Papers 25, 30, and 41 build the gauge structure. Papers 33 and 28 develop QED spectral sums on S^3 and the graph in a vertex model: its selection-rule census and its perturbative period content. Paper 2 reports the α observation. Paper 36 turns the Dirac-S^3 machinery onto precision bound-state observables. Paper 51 sketches the spectral-action gravity sector and—characteristically—records what did not work.

II. GAUGE STRUCTURE FROM THE HOPF BUNDLE

a. The Hodge/Wilson dictionary on the graph (Paper 25). With B the node–edge incidence matrix of the GeoVac graph, the node Laplacian $L_0 = BB^\top$ is the matter propagator of the lattice gauge theory (it does *not* converge to $-\Delta_{S^3}$: it splits into $n_{\max}$ disconnected ℓ-blocks, and the Coulomb spectrum lives on the labels instead [3]) and the edge Laplacian $L_1 = B^\top B$ is the discrete Hodge-1 Laplacian whose kernel counts cycles—the photon propagator in temporal gauge. Assigning a $U(1)$ connection $U_e = e^{i\theta_e}$ to each edge and collapsing the magnetic label m_ℓ over its $2\ell+1$ rotation multiplet, a stand-in for the Hopf fibre (whose own labels mix ℓ within a shell), gives the S^2 base graph; grouping by the fibre's own chiral factor instead (measured, pre-registered) gives one connected base with an irrational spectrum, so the three components belong to the rotation grouping. None of the underlying mathematics is new (graph Hodge theory, Wilson lattice gauge theory, the Marcolli–van Suijlekom gauge-network spectral triple [16]); what Paper 25 claims is the *synthesis*—one physically motivated discretization on which matter, an abelian connection, and the Hopf base coexist. The paper is explicit about its tier: "an observational synthesis paper. No new experimental or computational prediction is derived," and the edge-Laplacian spectrum of a finite Hopf graph is necessarily algebraic (π-free), so no transcendental can hide in the gauge 1-forms themselves.

b. $\mathrm{SU}(2)$ *because* $S^3 = \mathrm{SU}(2)$ *(Paper 30).* The rank-1 non-abelian case is special: the manifold *is* the gauge group. With links $U_e \in \mathrm{SU}(2)$, plaquettes the primitive non-backtracking (Ihara) cycles, and Wilson action $S_W = \beta \sum_P \big(1 - \frac{1}{2}\,\mathrm{Re}\,\mathrm{tr}\,U_P\big)$, the construction is a gauge-invariant, Haar-normalizable finite theory. Two results carry real weight. The maximal-torus (Cartan) reduction reproduces Paper 25's $U(1)$ action to machine precision ($|S_W - S_{U(1)}| < 10^{-12}$)—stated as a **proposition with a proof**—so Paper 25 is exactly the abelian sub-sector of a genuinely non-abelian theory. The second-order weak-coupling expansion produces the co-exact (curl) plaquette form $B_2 B_2^\top$ — *not* a multiple of L_1; it is the missing half that completes Paper 25's $L_1 = B^\top B$ to the full discrete Hodge-1 Laplacian (Paper 30 Prop. 2, corrected 2026-07-03) — with non-abelian couplings first entering at order A^4. The kinetic normalization is universal,

$$\text{coeff} = \frac{1}{4N_c} \implies \tfrac{1}{8}\ (\mathrm{SU}(2)), \quad \tfrac{1}{12}\ (\mathrm{SU}(3)), \qquad (1)$$

verified symbolically and numerically. The scope is explicitly bounded: this is a compact-graph construction, *not* a Yang–Mills mass-gap or confinement claim on $\mathbb{R}^4$; on this graph the plaquettes form a cycle basis of each planar l-block, so Wilson loops factorize and $\langle W \rangle = (I_2/I_1)^{\mathrm{Area}}$ exactly, which long Monte Carlo chains reproduce.

c. Where the gauge reach stops. The non-abelian story does not continue to SU(3) as a *natural* matter-coupled theory. On the S^5 Bargmann–Segal lattice (the harmonic-oscillator analog of S^3) a pure-gauge Wilson SU(3) is well defined and machine-precision verified (the same $1/(4N_c) = 1/12$ coefficient, Cartan reduction over 7,765 plaquettes to $< 1.5\times10^{-11}$), but the natural matter sector cannot be coupled in the strict Wilson sense: the Clebsch–Gordan intertwiner obstruction re-appears at the matter level, and a least-squares fit of the m_ℓ-quotient to $\mathbb{CP}^2$ fails uniformly (40.8% least-squares residual, with a convention-robust $\geq 33\%$ floor; Paper 25 §VII). This is the gauge-side face of the Coulomb/HO asymmetry catalogued in Paper 24 [5]: $U(1)$ transfers universally, SU(2) is the natural choice because $S^3 = \mathrm{SU}(2)$, and SU(3) admits a pure-gauge Wilson theory but no natural matter coupling. The cumulative reading is that the natural gauge content of the construction saturates at $U(1)\times\mathrm{SU}(2)\times\mathrm{SU}(3)$—the structural reach, not a derivation of the Standard-Model gauge group, and consistent with the spectral-triple gauge analysis of Paper 32 [9].

d. The Dirac Rule-B graph and 3D compact $U(1)$ *(Paper 41).* Paper 25's scalar Wilson $U(1)$ has a structural limitation: on the scalar Fock graph the gauge theory factorizes into disjoint per-ℓ two-dimensional grids, so it is really 2D compact $U(1)$ with no genuine three-dimensional phase structure. The cross-ℓ Dirac "Rule B" graph (the $E1$ dipole adjacency, $|\Delta\ell| = 1$) repairs this: every edge flips ℓ-parity, the graph is a single connected component, and at fixed diffusion scale its spectral dimension sits about 0.45 above the scalar graph's (an earlier "climb toward 3" was an estimator effect). Wilson $U(1)$ on Rule B is then tested against seven diagnostics (spectral dimension, Migdal–Kadanoff RG flow, Gaussian perimeter law, strong-coupling area law, a DeGrand–Toussaint monopole density decaying over 3.7 decades, full Monte-Carlo Wilson loops, and finite-T Polyakov loops on $G_B \times C_{N_t}$). The verdict is compatibility with

Construction	Count	Reading
Scalar graph + scalar photon	1/8	graph-native (Gaunt)
Native Dirac graph (scalar photon)	4/8	three by edge definition
Vector photon + scalar electrons	7/8	vertex model, $1/(4\pi)$/loop
Vector photon + Dirac electrons	8/8	vertex model, α_z only

TABLE I. Paper 33's census of eight selection rules (five standard, three specific to S^3) in a vertex model. The word "scalar" is overloaded: the 1/8 row is the scalar *photon* on the scalar graph; the 7/8 row is scalar *electrons* with the *vector* photon.

compact $U(1)$ and no 3D-versus-4D discrimination: no 4D control was run, the monopole data end below the 4D transition, the full-MC area coefficient is consistent with zero, and the Polyakov-loop confinement reading is withdrawn ($\langle P\rangle = 0$ is forced by centre symmetry).

III. QED ON S^3 AND THE GRAPH: SELECTION RULES AND PERIODS

The QED arc asks how much of perturbative quantum electrodynamics is visible on the combinatorial graph, before any continuum projection, and what period content the continuum spectral sums carry.

a. The selection-rule census (Paper 33). On the scalar Fock graph with a scalar (1-cochain) photon exactly one of the eight rules survives—the Gaunt/Clebsch–Gordan angular sparsity of the discrete edge set. Promoting the photon to a vector harmonic with labels (q, m_q) and Wigner-3j coupling scores six more, with the vertex normalization contributing $1/(4\pi)$ per loop (read as the S^2 Weyl exchange constant, an Observation). These scores are within a vertex model: the ground-state rule is the zero of a scalar-harmonic triangle rule, which the physical vector-photon vertex (allowing $q = n_1 + n_2 + 1$) does not enforce, and the eighth rule for Dirac electrons rests on a diagonal α_z identity that fails for a physical vector photon, whose diagonal magnetic coupling is nonzero. The census with the physical vertex is open.

b. The perturbative period tower (Paper 28). The S^3 Dirac spectral sums (Camporesi–Higuchi spectrum $|\lambda_n| = n + \frac{3}{2}$, Hodge-1 photon, the vertex model's selection rule) produce odd zeta values and the S^3 vertex-parity content G, $\beta(4)$ through a spectral mechanism organized by operator order and vertex topology. The spectral-zeta identities are statements about the sums as defined and stand independently of the vertex model:

- **T9 (two-term theorem).** The D^2 spectral zeta is a two-term π^{even} polynomial with rational coefficients at every integer s—no odd-zeta content there (the one-loop determinant $\zeta'_{D^2}(0) = 3\zeta_R(3)/(4\pi^2) + \frac{1}{2}\ln 2$ does carry it); operator order is the primary transcendental discriminant, conditional on the period conjecture.
- **χ_{-4} bridge (Theorem 3).** A spectral-zeta-to-Dirichlet-L identity, $D_{\text{even}}(s) - D_{\text{odd}}(s) = 2^{s-1}\big(\beta(s) - \beta(s-2)\big)$ with $\beta(s) = L(s, \chi_{-4})$, in which vertex parity acts as the Dirichlet character; proven, and verified symbolically and by PSLQ to 100 digits. It is consistent with but does not enforce the Riemann hypothesis for β.
- **Realized-depth tower.** Loop order bounds the multiple-zeta depth but does not automatically mint new irreducibles; the realized depth grows strictly more slowly, $\leq k - 1$. The two-loop sunset constant reduces entirely, $S_{\min} \in \mathbb{Q}[\pi^2, \ln 2, \zeta(3), \zeta(5)]$ (depth ≤ 1); the three-loop chain $S^{(3)} = 31.5726\ldots$ first introduces a depth-2 period, $\zeta(5,3)$, as its only constant beyond the single-zeta ring (its irreducibility to single zetas is conjectural).

On the finite graph these quantities are π-free rationals or algebraic numbers—the transcendentals enter only through the infinite spectral sums of the continuum projection. The vertex model's ground-state self-energy zero is not reproduced on the graph (the pendant-edge proposition gives $\Sigma_{\text{graph}}(\text{GS}) = 2(n_{\max} - 1)/n_{\max} \to 2$), and the plaquette transverse propagator's zero holds for every s-state (each s-edge is a bridge), so it recovers no ground-state rule. A spectral-sum anomalous-moment ratio of 1.084 divides a coupling-free sum by the CODATA $\alpha/(2\pi)$; its closeness to $1 + 2/25$ is a numerical coincidence.

IV. THE FINE-STRUCTURE-CONSTANT OBSERVATION

The arc's most precise—and most carefully bounded—result is Paper 2's. The fine-structure constant approximately satisfies (to 8.8×10^{-8} relative; an Observation) the depressed cubic

$$\alpha^3 - K\alpha + 1 = 0, \qquad K = \pi(B + F - \Delta) = \pi\left(42 + \frac{\pi^2}{6} - \frac{1}{40}\right), \tag{2}$$

where the three coefficients are spectral invariants of the Hopf bundle components: $B = 42$ is the degeneracy-weighted Casimir trace of the S^2 base (with the selection principle $B/N = \dim S^3 = 3$ proven algebraically to fix the cutoff $n_{\max} = 3$ uniquely), $F = \zeta(2) = \pi^2/6$ is half the spectral zeta of the S^1 fiber at $s = 1$, its $k > 0$ modes (equivalently the Fock-degeneracy Dirichlet series at the packing exponent $d_{\max} = 4$ [1]), and $\Delta = 1/40$ is an S^3 boundary term equal to the reciprocal of the single-level Dirac degeneracy $g_3^{\text{Dirac}} = 2(n+1)(n+2)|_{n=3} = 40$. The physical root gives $\alpha^{-1} = 137.036011$, matching CODATA to a relative error of 8.8×10^{-8} with zero free parameters; in a combinatorial search over 1.92×10^9 comparable formulas the formula ranks among the top ~ 10 ($p = 5.2 \times 10^{-9}$ per draw), a statistic that does not

account for the unregistered search that produced the formula and does not establish non-accidentality.

The status of Eq. (2) must be stated exactly, and it is a **hard prohibition** of the framework that it be stated no more strongly. The combination rule $K = \pi(B + F - \Delta)$ is an **Observation**: an empirical numerical coincidence with structural support, *never* a derivation, conjecture, prediction, or theorem. Each ingredient has an *identified* spectral home—a "three-homes" theorem establishes that B, F, Δ live in two distinct sectors of S^3 (scalar and spinor) and are three different object types (finite trace, infinite Dirichlet series, single-level degeneracy); no single construction generating all three has been exhibited (an empirical record, not part of the theorem)—but their *combination* is not derived. A multi-phase structural-decomposition program eliminated twelve single-mechanism derivations of the rule, leaving the cross-sector sum a genuine open question: in the paper's own assessment, the geometric arena (Link 1) and the per-component invariants (Link 2) are largely established, while the additive combination (Link 3) is "not established." The synthesis inherits this verdict verbatim: what is structural here is each spectral home; what is observational is that they sum approximately to α^{-1}/π (to 4.8×10^{-7} relative; 8.8×10^{-8} through the cubic's root).

V. BOUND-STATE PRECISION QED

Paper 36 turns the Dirac-S^3 infrastructure onto Lamb-shift-class observables, where the answer is known independently and the test is one of internal consistency rather than prediction. The hydrogen $2S_{1/2}$–$2P_{1/2}$ Lamb shift at leading one-loop order, assembled in the Eides convention with the tabulated Drake–Swainson Bethe logarithms, is 1052.19 MHz against the experimental 1057.845 MHz (-0.534%). That headline is textbook one-loop arithmetic with literature inputs, and it omits the one-loop $\alpha(Z\alpha)^5$ term (about +6.9 MHz), so the residual is not attributed. The GeoVac content is the Bethe-logarithm machinery (Sturmian basis at $\lambda = Z/n$, acceleration form, Drake–Swainson asymptotic subtraction, which is Paper 34's thirteenth catalogued projection [11]): it reproduces the tabulated $2P$ and $3D$ logarithms to 2.4% and 0.07%, while the $2S$ acceleration-form value is a crossing point, not yet converged. The leading one-loop assembly transfers to muonic hydrogen under the rest-mass projection; a muonium comparison is withdrawn because its theory reference could not be located in the cited source. The two-loop sector is the explicit boundary: the bare iterated spectral action reproduces the flat-space two-loop prefactor structure but does not generate renormalization counterterms, so finite multi-loop predictions remain external input.

VI. THE SPECTRAL-ACTION GRAVITY SKETCH

The most speculative paper in the arc, Paper 51, asks whether the Fock-projected S^3 supports a sensible gravity reading at the Chamseddine–Connes spectral-action level [17]. It is positioned, correctly and throughout, as an *exploratory, structural-sketch* program—"with the standard caveats of the Connes–Chamseddine program"—not as established physics, and it is unusually candid about what failed.

One result is solid, and its content is Chamseddine and Connes's: the spectral action on the round S^3 (and $S^3 \times S^1$) is two-term up to exponentially small corrections—Einstein–Hilbert plus a cosmological constant, with no higher-curvature power corrections at any order. Paper 51 gives an alternative algebraic proof through the spectral-zeta identity $\zeta_{\text{unit}}(-k) = 0$ for all $k \geq 0$ (via the Bernoulli identity $B_{2k+1}(3/2) = (2k+1)/4^k$); it is spinor-bundle-specific (the scalar and transverse-traceless sectors retain the full infinite Seeley–DeWitt series). From it follow, at the spectral-action level, a Bekenstein–Hawking entropy $S_{\text{BH}} = A\Lambda^2/(12\pi)$ (continuum replica derivation), an effective Newton constant $G_{\text{eff}} = 3\pi/\Lambda^2$ (rank-4 Dirac trace), a formal de Sitter extremum of the two-term form (the exact action is monotonic), a proven Fierz–Pauli "J-blindness" theorem, and a proven (Layer-2, rate-level) replica-weight-harmless theorem.

The negatives are the point, and they are stated plainly:

- the per-t "UV target" $1/(24\pi t)$ once used to grade the discrete tip extraction was dimensionally wrong: the continuum cone tip term is t-independent (1/6), and the discrete per-t recovery at the lattice scale fails in both directions against it;
- the Möbius factor $\alpha/(2\alpha-1)$, once read as a continuum $\alpha > 1$ "mechanism," is a *finite-lattice-spacing artifact*: as the radial mesh refines, the recovery climbs back toward the plain Sommerfeld–Cheeger value, so there is no exotic mechanism to find;
- no cutoff function cures the infrared over-count across all Λ (clean negative);
- the Newton-constant "factor of 2" between the action and entropy routes was an error, not a convention: the action route had been traced over a rank-2 bundle, and the rank-4 Dirac trace makes the two agree; it was never a scalar-vs-Dirac cone-coefficient difference (2D Dirac and scalar cones share the magnitude $(1/12)|1/\alpha - \alpha|$, with opposite signs);
- Bisognano–Wichmann wedge entanglement entropy is *not* an area law—it scales as $\sim 2 \log n_{\max}$ (area-law fit rejected at $R^2 = 0.83$), so the entropy

comes from the replica/spectral-action route, not from wedge entanglement.

The overall verdict is that the round-sphere spectral action supplies a two-term Einstein–Hilbert skeleton (Chamseddine–Connes; an alternative proof here), the cutoff function remains external calibration data, and the full discrete-substrate re-derivation of S_{BH} is an explicitly open research program.

VII. HONEST SCOPE AND POSITIONING

a. What is robust. The load-bearing results are structural and proven. The Hodge/Wilson dictionary ($L_0 = BB^\top$ matter, $L_1 = B^\top B$ photon) is exact, as is the machine-precision SU(2) maximal-torus reduction that makes Paper 25 the Cartan sector of Paper 30, and Wilson loops on the scalar graph factorize exactly. The perturbative period results—the T9 two-term theorem, the χ_{-4} Dirichlet-L bridge, and the realized-depth tower with $\zeta(5,3)$ as the first depth-2 period (conjecturally irreducible)—are theorems (T9, χ_{-4}) and a Proposition (the realized-depth tower, PSLQ-identified) about the sums as defined. The spectral-action two-term exactness on S^3 is a theorem (Chamseddine–Connes; alternative proof in Paper 51). Each of the three α-ingredients B, F, Δ has an identified spectral home.

b. What is conditional or observational. The fine-structure combination rule $K = \pi(B + F - \Delta)$ is an **Observation**, not a derivation—the single most important tier statement in the arc. The selection-rule census is a statement about a vertex model, not about the physical QED vertex. The bound-state QED headline is standard one-loop arithmetic; the GeoVac content is its Bethe logarithms. The Rule-B gauge theory is compatible with compact $U(1)$ with no 3D-versus-4D discrimination. The entire gravity sector is a spectral-action-level structural sketch with an open discrete-substrate program.

c. What was tried and did not work. The arc is bounded by clean negatives. SU(3) has no natural matter-coupled lattice gauge theory—the Clebsch–Gordan intertwiner obstruction caps the natural gauge reach (a face of the Coulomb/HO asymmetry). The scalar graph photon misses seven of the eight selection rules because it carries no angular quantum numbers. The vertex model's ground-state zero, the plaquette "recovery" of it, the α_z Furry identity, the anomalous-moment ratio's Parker–Toms match, the Lamb-shift residual decomposition, and the Rule-B Polyakov-loop confinement reading are withdrawn or re-scoped. Twelve single-mechanism derivations of the α combination rule were eliminated. The gravity sector contributes five recorded negatives—the Möbius finite-spacing artifact, the failed IR-over-count cure, the withdrawn per-t tip target, the factor of 2 traced to the spinor rank, and the non-area-law wedge entropy—each kept on the record so it is not re-derived.

d. Positioning. Read as a whole, the Group 5 arc is a structural map of how far the spectral geometry of S^3 reaches into electromagnetism, the gauge sector, and gravity, with the boundary between forced structure and free coupling drawn explicitly at every step. The natural gauge constructions, the QED period tower of the spectral sums, and the two-term round-sphere action are the structural assets; the fine-structure constant, the QED multi-loop constants, and the gravity cutoff are the calibration side. The discrete and continuous descriptions are dual, with no ontological priority asserted for either. The arc is not a derivation of α, not a derivation of the Standard-Model gauge group, and not a theory of quantum gravity; it is a disciplined account of which structures the framework forces and which numbers it must be told.

[1] J. Loutey, "Counting Angular Momentum Labels by Isotropic Packing," GeoVac Paper 0 (2026).

[2] J. Loutey, "The Fine Structure Constant as a Spectral Coincidence on the Hopf Fibration," GeoVac Paper 2 (2026). [The combination rule $K = \pi(B + F - \Delta)$ is recorded strictly as an Observation; see §IV.]

[3] J. Loutey, "The Dimensionless Vacuum: Recovering the Schrödinger Equation on a Scale-Invariant Label Lattice," GeoVac Paper 7 (2026).

[4] J. Loutey, "Spectral-Geometric Exchange Constants: Projecting Discrete Spectra onto Continuous Manifolds," GeoVac Paper 18 (2026).

[5] J. Loutey, "The Bargmann–Segal Lattice: π-Free Discretization of the Harmonic Oscillator on S^5," GeoVac Paper 24 (2026).

[6] J. Loutey, "The Hopf Graph as a Lattice Gauge Structure: A Framework Observation," GeoVac Paper 25 (2026).

[7] J. Loutey, "QED on S^3: Spectral Geometry of Perturbative Transcendentals," GeoVac Paper 28 (2026).

[8] J. Loutey, "SU(2) Wilson Lattice Gauge Structure on the GeoVac Hopf Graph: The Non-Abelian Sibling of Paper 25," GeoVac Paper 30 (2026).

[9] J. Loutey, "The GeoVac Spectral Triple," GeoVac Paper 32 (2026).

[10] J. Loutey, "The 1+6+1 Partition of QED Selection Rules on S^3: Graph Topology, Angular Momentum, and the Diagonal α_z Vertex," GeoVac Paper 33 (2026).

[11] J. Loutey, "GeoVac as a Two-Layer Framework: Projection Taxonomy and the Empirical Matches Catalog," GeoVac Paper 34 (2026).

[12] J. Loutey, "Bound-State QED on S^3: The Leading One-Loop Hydrogen Lamb Shift," GeoVac Paper 36 (2026).

[13] J. Loutey, "State-space Gromov–Hausdorff convergence of truncated Camporesi–Higuchi spectral triples on S^3," GeoVac Paper 38 (2026).

[14] J. Loutey, "Wilson $U(1)$ Lattice Gauge on the Dirac-Rule-B Graph: Compatibility with Compact $U(1)$ on a Non-Cubic Substrate, Including Monopole-Density and Polyakov-Loop Diagnostics," GeoVac Paper 41 (2026).

[15] J. Loutey, "Gravity from the GeoVac Spectral Action: Continuum Results and the Discrete-Substrate Program," GeoVac Paper 51 (2026).

[16] M. Marcolli and W. D. van Suijlekom, "Gauge networks in noncommutative geometry," J. Geom. Phys. **75**, 71 (2014); arXiv:1301.3480.

[17] A. H. Chamseddine and A. Connes, "The Spectral Action Principle," Commun. Math. Phys. **186**, 731 (1997).